Bangladesh University of Business and Technology



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Lab Report

Experiment No : 01
Course Title : Data Structure and Algorithm
Course Code : CSE 222
Experiment Name : Array insertion and deletion, Registration management

Submitted By

Name : Mst. Rabeya
Id No : 20261203020
Intake : 50th
Section : 01

Submitted To:

Name: Iffat Ara Sanzida

Dept. of CSE
Bangladesh University of Business and Technology

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Program-01: Array Insertion and Deletion

Source Code:

```
#include <iostream>
using namespace std;

int main()
{
    int arr[100];
    int n, choice, value, pos;

    cout << "Enter number of elements: ";
    cin >> n;

    cout << "Enter " << n << " elements: ";
    for (int i = 0; i < n; i++)
    {
        cin >> arr[i];
    }

    cout << "\n1. Insert";
    cout << "\n2. Delete";
    cout << "\nEnter your choice: ";
    cin >> choice;

    if (choice == 1)
```

```
{
    cout << "Enter value to insert: ";
    cin >> value;

    cout << "Enter index: ";
    cin >> pos;

    for (int i = n; i > pos; i--)
    {
        arr[i] = arr[i - 1];
    }

    arr[pos] = value;

    n++;

    cout << "Updated array: ";
    for (int i = 0; i < n; i++)
    {
        cout << arr[i] << " ";
    }
}

else if (choice == 2)
{
    cout << "Enter index to delete: ";
```

```
        cin >> pos;

        for (int i = pos; i < n - 1; i++)
        {
            arr[i] = arr[i + 1];
        }

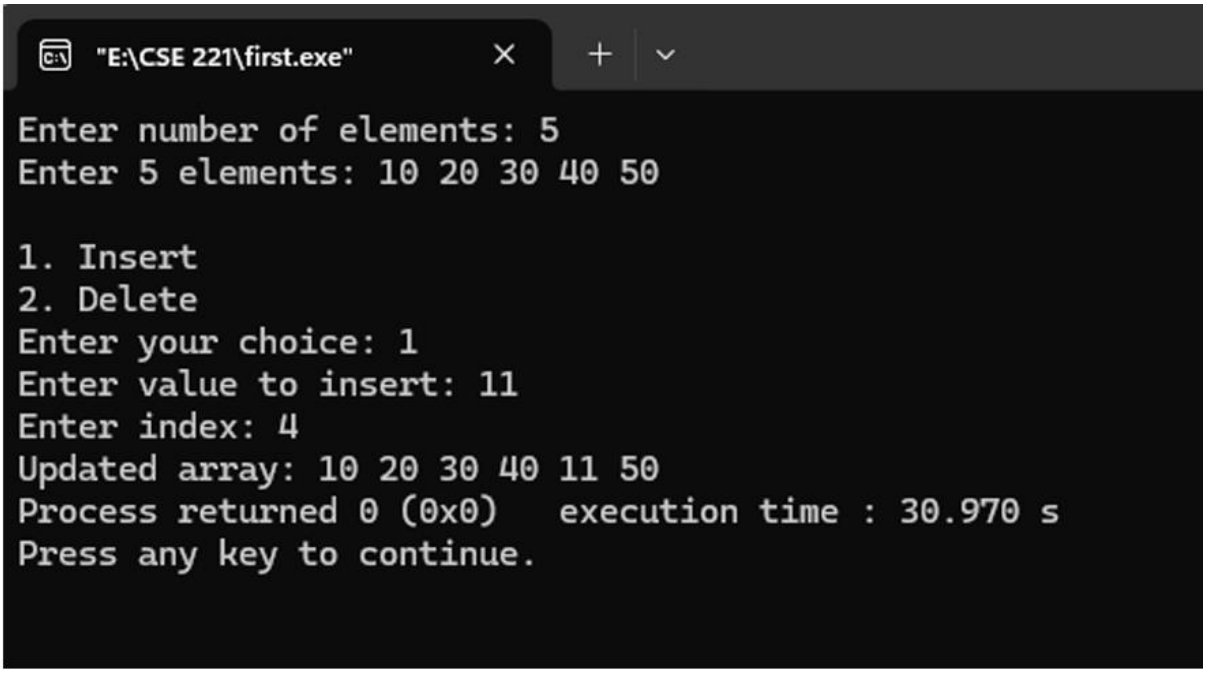
        n--;

        cout << "Updated array: ";
        for (int i = 0; i < n; i++)
        {
            cout << arr[i] << " ";
        }
    }

    else
    {
        cout << "Invalid choice!";
    }

    return 0;
}
```

Output:



Program-02: Registration management using Array.

Source Code:

```
#include <stdio.h>


int main()
{
    int course[100];
    int n, search, newCourse, pos;
    int found = 0;


    printf("Enter number of courses: ");
    scanf("%d", &n);


    printf("Enter course IDs:\n");
    for (int i = 0; i < n; i++)
```

```
{
    scanf("%d", &course[i]);
}

printf("\nRegistered courses: ");
for (int i = 0; i < n; i++)
{
    printf("%d ", course[i]);
}

printf("\n\nEnter course ID to search: ");
scanf("%d", &search);

for (int i = 0; i < n; i++)
{
    if (course[i] == search)
    {
        found = 1;

        for (int j = i; j < n - 1; j++)
        {
            course[j] = course[j + 1];
        }

        n--;
        printf("Course %d deleted.\n", search);
```

```
        break;
    }
}

if (found == 0)
{
    printf("Course not found.\n");
}

printf("\nEnter new course ID: ");
scanf("%d", &newCourse);

printf("Enter position: ");
scanf("%d", &pos);

for (int i = n; i > pos; i--)
{
    course[i] = course[i - 1];
}

course[pos] = newCourse;
n++;

printf("\nFinal course list: ");
for (int i = 0; i < n; i++)
{
```

```
        printf("%d ", course[i]);
    }

    return 0;
}
```

Output:

